



SCOPE OF ACCREDITATION TO ISO/IEC 17025:2005
& ANSI/NCSL Z540-1-1994

ADVANCED TECHNOLOGY SERVICES - CATERPILLAR PROVING GROUNDS

Peoria Proving Ground, 27194 Caterpillar Lane

Washington, IL 61571

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CALIBRATION

Valid until: September 30, 2018

Certificate Number: 1592.06

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations¹:

I. Dimensional

Parameter/Equipment	Range	CMC ^{2,3} (±)	Comments
Linear Displacement – Travel	(0 to 2032) mm	(5 + 0.16L) μm	Heidenhain linear encoder LS106
Angle	(0 to 60) °	0.16 °	Gage blocks w/ sine bar

II. Electrical – DC/Low Frequency

Parameter/Equipment	Range	CMC ^{2,5} (±)	Comments
DC Voltage – Generate	(0 to 220) mV 220 mV to 2.2 V (2.2 to 11) V (11 to 22) V (22 to 220) V (220 to 1000) V	8.9 μV/V + 0.40 μV 5.3 μV/V + 0.70 μV 3.8 μV/V + 2.5 μV 4.4 μV/V + 4.0 μV 5.3 μV/V + 40 μV 6.8 μV/V + 0.40 mV	Fluke 5730A

Parameter/Equipment	Range	CMC ^{2,5} (±)	Comments
DC Voltage – Measure	(0 to 100) mV	69 µV/V + 4.2 µV	Agilent 34410A
	100 mV to 1 V	42 µV/V + 9.7 µV	
	(1 to 10) V	36 µV/V + 60 µV	
	(10 to 100) V	49 µV/V + 0.72 mV	
	(100 to 1000) V	56 µV/V + 10 mV	
	5 V	0.75 mV	Somat cal stand/Keithley 2750/ MSBRG
	10 V	0.87 mV	
	(+20 to -20) mV	0.046 mV	
	(+10 to -10)	5.9 mV	
	(+2.5 to -2.5) V	1.8 mV	
	(+2 to -2) V	1.5 mV	
	(+5 to -5) V	2.7 mV	
	(-0.5 to 2) V	0.90 mV	
	(-2 to 0.5) V	0.94 mV	
	(+1 to -1) V	0.72 mV	
	(+200 to -200) mV	0.15 mV	
	(+250 to -250) mV	0.19 mV	
	(+500 to -500) mV	0.38 mV	
	(0 to 400) mV	0.15 mV	
	(375 to 500) mV	0.086 mV	
	3 V	24 mV	Somat cal stand/Keithley 2750/ MSHLS
	5 V	20 mV	
	10 V	19 mV	
	15 V	20 mV	
	20 V	20 mV	
	24 V	23 mV	
	28 V	33 mV	
	(-120 to +120) mV	0.13 mV	
	(-10 to +10) V	4.4 mV	
	(+60 to -60) mV	0.10 mV	
	(+80 to -80) V	45 mV	
	(+10 to -10) V	3.3 mV	
	(+2.5 to -2.5) V	0.97 mV	
	(+2 to -2) V	0.76 mV	
	(+5 to -5) V	2.0 mV	
	(0 to +10) V	3.0 mV	
	(-10 to 0) V	2.8 mV	
	(+1 to -1) V	0.51 mV	
	(+120 to -120) mV	0.12 mV	
	(0 to +120) mV	0.10 mV	
	(-120 to 0) mV	0.11 mV	
	(0 to 240) mV	0.12 mV	
	(-240 to 0) mV	0.13 mV	
	(+500 to -500) mV	0.26 mV	

Parameter/Equipment	Range	CMC ^{2,5} (±)	Comments
DC Voltage – Measure (cont.)	4 V	24 mV	SoMat cal stand/Kiethley 2750/ELHLS
	5 V	22 mV	
	9 V	25 mV	
	10 V	24 mV	
	12 V	23 mV	
	14 V	24 mV	
	15 V	24 mV	
	(+62.5 to -62.5) mV	0.045 mV	
	(+16 to -16) V	4.4 mV	
	(+10 to -10) V	2.2 mV	
	(+2.5 to -2.5) V	0.52 mV	
	(+2 to -2) V	0.45 mV	
	(+5 to -5) V	1.1 mV	
	(+1 to -1) V	0.24 mV	
	(+200 to -200) mV	33 µV	
	(-125 to +125) mV	27 µV	
	(-250 to +250) mV	53 µV	
	(-400 to +400) mV	83 µV	
	(+500 to -500) mV	91 µV	
	(-10 to +10) V	6.3 mV	SoMat cal stand/Kiethley 2750/MSHLB
	(-20 to +20) V	9.0 mV	
	(-100 to +100) mV	0.047 mV	SoMat cal Stand/Kiethley 2750/MSTCB
	(-100 to +100) mV	0.045 mV	SoMat cal stand/Kiethley 2750/ELTCB
	5 V	0.19 mV	SoMat cal stand/Kiethley 2750/SMSTRB-120
	10 V	0.37 mV	
	10 Gain	48 µV	
	100 Gain	54 µV	
	5 V	0.16 mV	SoMat cal stand/Kiethley 2750/SMSTRB-350
	10 V	0.30 mV	
	10 Gain	0.046 mV	
	100 Gain	0.34 mV	

Parameter/Equipment	Range	CMC ^{2,5} ($\pm$)	Comments
Resistance – Measure	(0 to 100) Ω 100 Ω to 1 k Ω (1 to 10) k Ω (10 to 100) k Ω 100 k Ω to 1 M Ω (1 to 10) M Ω 50 k Ω 100 k Ω 200 k Ω 500 k Ω	0.15 m Ω/Ω + 4.8 m Ω 0.12 m Ω/Ω + 12 m Ω 0.12 m Ω/Ω + 12 m Ω 0.12 m Ω/Ω + 1.2 Ω 0.14 m Ω/Ω + 12 Ω 0.49 m Ω/Ω + 0.12 k Ω 12 Ω 19 Ω 240 Ω 610 Ω	Agilent 34410A Somat cal stand/Keithley 2750/ MSBRG
Thermocouple Simulation – Generate Type J Type K Type T Type E	(-210 to 760) $^{\circ}\text{C}$ (-140 to 800) $^{\circ}\text{C}$ (800 to 1372) $^{\circ}\text{C}$ (-200 to -100) $^{\circ}\text{C}$ (-100 to 400) $^{\circ}\text{C}$ (-270 to -260) $^{\circ}\text{C}$ (-260 to -250) $^{\circ}\text{C}$ (-250 to -240) $^{\circ}\text{C}$ (-240 to -230) $^{\circ}\text{C}$ (-230 to 1000) $^{\circ}\text{C}$	0.31 $^{\circ}\text{C}$ 0.27 $^{\circ}\text{C}$ 0.29 $^{\circ}\text{C}$ 0.29 $^{\circ}\text{C}$ 0.26 $^{\circ}\text{C}$ 0.82 $^{\circ}\text{C}$ 0.47 $^{\circ}\text{C}$ 0.39 $^{\circ}\text{C}$ 0.33 $^{\circ}\text{C}$ 0.29 $^{\circ}\text{C}$	Xitron 2000M
Thermocouple Simulation – Measure	Type E Type J Type K Type T	1 $^{\circ}\text{C}$ (5.4 E-01) $^{\circ}\text{C}$ (7.4 E-01) $^{\circ}\text{C}$ (7.8 E-01) $^{\circ}\text{C}$	SoMat cal stand/Keithley 2750/SMITC

III. Mechanical

Parameter/Equipment	Range	CMC ^{2,4} (±)	Comments
Force	Up to 966 N (0 to 2224.1) N (0 to 8896.4) N (0 to 22241.1) N (0 to 1200) kN	0.19 N 0.035 % + 5.9 N 0.082 % + 22 N 0.070 % + 52 N 0.65 % + 0.16 kN	Class “F” weights ATS press & load cells Instron press & load cells
Torque	(0 to 10000) N·m	22 N·m + 0.0021 N·m	Drive shaft calibrator
Micro-Strain Indicators	(0 to 20 000) µE	1.9 µE + 280 µE/E	Vishay 1550A
Pressure – Hydraulic Pneumatic	0 kPa to 12 MPa (12 to 40) MPa 100 kPa to 42 MPa (42 to 140) MPa (0 to 206.84) kPa (206.84 to 689.8) kPa (100 to 2068.4) kPa (2068.4 to 6894.8) kPa	6.2 kPa + 0.0047 kPa 2.6 kPa + 0.16 kPa 6.3 kPa + 0.030 kPa 8.4 kPa + 0.17 kPa 7.0 kPa + 0.065 kPa 5.1 kPa + 0.077 kPa 6.9 % + 0.061 kPa 0.010 % + 0.42 kPa	Fluke PPCH DH Instruments PPC3
Vibration (Accelerometer)	(10 to 99) Hz 100 Hz (101 to 920) Hz (921 to 5000) Hz 5001 Hz to 10 kHz (10 to 15) kHz (15 to 20) kHz	2.2 % 1.9 % 2 % 2.4 % 2.8 % 4.5 % 5.4 %	Modal shop 9155C

¹ This laboratory offers commercial calibration service.

² Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of $k = 2$. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

³ In the statement of CMC, L is the numerical value of the nominal length of the device measured in millimeters.

⁴ In the statement of CMC, % is the percent of reading unless otherwise indicated.

⁵ The stated measured values are determined using the indicated instrument (see Comments). This capability is suitable for the calibration of the devices intended to measure or generate the measured value in the ranges indicated. CMC's are expressed as either a specific value that covers the full range or as a percent or fraction of the reading plus a fixed floor specification.





Accredited Laboratory

A2LA has accredited

ADVANCED TECHNOLOGY SERVICES, INC. - CATERPILLAR PROVING GROUNDS

Washington, IL

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets the requirements of ANSI/NCSL Z540-1-1994 and R205 – *Specific Requirements: Calibration Laboratory Accreditation Program*. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009).



Presented this 21st day of November 2016.

A handwritten signature in black ink, appearing to read 'L. L. L.', written over a horizontal line.

President and CEO
For the Accreditation Council
Certificate Number 1592.06
Valid to September 30, 2018

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.